

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026

Chemistry — Class IX

Syllabus: Ch 1 — Nature of Chemistry | Ch 2 — Matter | Ch 3 — Atomic Structure | Ch 4 — Periodic Table & Periodicity | Ch 5 — Chemical Bonding | Ch 6 — Stoichiometry | Ch 7 — Electrochemistry | Ch 8 — Energetics | Ch 9 — Chemical Equilibrium

Total Marks: 65

Time Allowed: 2 Hours 30 Minutes

Version: 6

General Instructions:

1. This paper has three sections: Section A (MCQs), Section B (Short Questions), and Section C (Long Questions).
2. Attempt Section A on the question paper by filling the bubble next to the correct option.
3. In Section B, attempt all 11 questions. For each, choose either the main question OR its alternative.
4. In Section C, attempt all 4 questions. For each, choose either the main question OR its alternative. Show all working; marks are awarded for method, not only the final answer.
5. Atomic Masses (amu): H=1, C=12, N=14, O=16, Na=23, Al=27, P=31, S=32, Cl=35.5, K=39, Ca=40, Cu=63.5 — Avogadro's Number $N_A = 6.022 \times 10^{23}$

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Question	A	B	C	D	Answer
1	Which branch of chemistry improves techniques to gauge the behaviour of pollutants and develops methods for pollution control?	Organic Chemistry	Analytical Chemistry	Environmental Chemistry	Geochemistry	ⒶⒷⒸⒹ
2	According to the solubility curves, which of the following shows DECREASING solubility as temperature rises from 0°C to 100°C?	KNO ₃	NH ₄ Cl	SO ₂	NaCl	ⒶⒷⒸⒹ
3	An atom is represented as $^{27}_{13}\text{M}$. The number of neutrons in this atom is:	13	14	27	40	ⒶⒷⒸⒹ
4	Which of the following statements is NOT correct about isotopes of an element?	Same atomic number	Same number of protons	Same chemical properties	Same physical properties	ⒶⒷⒸⒹ
5	For normal elements, valence electrons = group number. An atom of $^{32}_{16}\text{S}$ has how many valence electrons?	2	4	6	16	ⒶⒷⒸⒹ
6	Which of the following elements would be expected to have the greatest electron affinity?	S	P	Cl	Na	ⒶⒷⒸⒹ
7	Silicon belongs to Group IVA of the periodic table. How many electrons does it have in its valence shell?	2	3	4	6	ⒶⒷⒸⒹ
8	Which of the following atoms (given atomic number) will NOT form a cation or an anion?	Atomic No. 16	Atomic No. 17	Atomic No. 18	Atomic No. 19	ⒶⒷⒸⒹ
9	What is the mass of carbon present in 44 g of carbon dioxide (CO ₂)? (C=12, O=16)	44 g	32 g	12 g	6 g	ⒶⒷⒸⒹ
10	Which term is the same for one mole of oxygen gas (O ₂) and one mole of water (H ₂ O)?	Mass	Volume	Number of molecules	Number of atoms	ⒶⒷⒸⒹ
11	Air pollutants SO ₂ and NO ₂ combine with oxygen and water vapour to form acid rain. Which two acids are formed?	HCl & H ₂ CO ₃	H ₂ SO ₄ & HNO ₃	H ₂ SO ₃ & HNO ₂	H ₂ S & HNO ₃	ⒶⒷⒸⒹ
12	In the reaction of limestone with hydrochloric acid solution in a test tube, the "system" refers to:	The test tube only	Limestone and HCl solution	The room around the tube	The heat source	ⒶⒷⒸⒹ

SECTION B — Short Questions (11 × 3 = 33 Marks)

Attempt all 11 questions. For each pair, attempt either the main question OR its alternative.

Part	Question	Marks	OR	OR Question	Marks
2(i)	How does forensic chemistry help us in solving crimes? Name the broader branch of chemistry to which it belongs.	3	OR	The batteries used in vehicles are built on the principle of electrochemistry. Which branch of chemistry is this an application of? Give one other example of this branch's application in daily life.	3
2(ii)	Distinguish between a homogeneous solution and a heterogeneous solution, giving one example of each.	3	OR	Why is a solution considered a mixture rather than a compound? How would you test whether a given liquid sample is a colloid or a true solution?	3
2(iii)	Distinguish between a shell and a sub-shell. How many sub-shells are present in the N-shell (n=4)? Name them.	3	OR	Give the notation for the sub-shells present in the M-shell (n=3). Arrange them in order of increasing energy.	3

Part	Question	Marks	OR	OR Question	Marks
2(iv)	State which sub-shell can accommodate a maximum of 6 electrons. Write the electronic configuration of sodium (Z=11) using s,p notation (e.g. $1s^2 2s^2 \dots$).	3	OR	Can an atom be identified without knowing the number of neutrons in it? Explain your answer with reference to atomic number.	3
2(v)	Which atom has a higher shielding effect, lithium or sodium? Explain why.	3	OR	Explain why sodium has a higher ionization energy than potassium.	3
2(vi)	For normal elements, the number of valence electrons equals the group number. Determine the group number of aluminium ($^{27}_{13}\text{Al}$) and potassium ($^{39}_{19}\text{K}$) from their electron configuration.	3	OR	Write the valence-shell electronic configuration (general form) for: (a) alkali metals (b) alkaline earth metals (c) halogens (d) noble gases.	3
2(vii)	State the octet rule and the duplet rule. Among oxygen (O), fluorine (F), hydrogen (H), and nitrogen (N), which one obeys the duplet rule instead of the octet rule? Explain why.	3	OR	Phosphorus belongs to Group VA of Period 3. How many electrons does it need to complete its valence-shell octet? Show the electron-dot structure of a phosphorus atom (Z=15).	3
2(viii)	Which of the following statements about the formation of the ionic compound Na_2S is NOT true: (a) each sodium atom loses one electron (b) sodium forms a cation (c) sulphur forms an anion (d) each sulphur atom gains one electron? Identify the incorrect statement and describe the correct formation of Na_2S .	3	OR	An atom with atomic number 8 will form an ion of charge -2 , while an atom with atomic number 12 will form an ion of charge $+2$. Explain both, using electron configuration, and name the ions formed.	3
2(ix)	Calculate the number of molecules present in 1.09 moles of benzene (C_6H_6). ($N_A = 6.022 \times 10^{23}$)	3	OR	Calculate the number of molecules present in 0.01 moles of acetic acid (CH_3COOH). ($N_A = 6.022 \times 10^{23}$)	3
2(x)	Calculate the mass of 1.05 moles of hydrated copper(II) sulphate, $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$. (Cu=63.5, S=32, O=16, H=1)	3	OR	Calculate the number of molecules present in 9.0 g of steam (H_2O). (H=1, O=16, $N_A = 6.022 \times 10^{23}$)	3
2(xi)	Explain how sulphur dioxide (SO_2) and nitrogen dioxide (NO_2), released as air pollutants, lead to the formation of acid rain. Name the two acids formed.	3	OR	Sodium nitrite (NaNO_2) is used as a food preservative. Determine the oxidation state of nitrogen in NaNO_2 , showing your working.	3

SECTION C — Long Questions ($4 \times 5 = 20$ Marks) — Attempt all 4

Part	Question	Marks	OR	OR Question	Marks
3(i)	(a) An atom has atomic number 15 and mass number 31. Calculate the number of protons, neutrons, and electrons it contains. [2] (b) State how many sub-shells are present in the N-shell ($n=4$), and give their notations. Which sub-shell can hold a maximum of 14 electrons? [3]	5	OR	(a) An atom has 20 protons and a mass number of 40. Determine its number of neutrons and electrons. [2] (b) Write the electronic configuration of this atom in s,p notation. State its group and period in the periodic table, and identify whether it is a metal or a non-metal. [3]	5
3(ii)	The valence-shell electronic configurations of eight elements are: $A = 1s^2 2s^2$ $B = 1s^2 2s^2 2p^6$ $C = 1s^2 2s^2 2p^3$ $D = 1s^2$ $E = 1s^2 2s^2 2p^6 3s^2 3p^3$ $F = 1s^2 2s^1$ $G = 1s^2 2s^2 2p^6 3s^1$ $H = 1s^2 2s^2 2p^6 3s^2$ (a) Group these elements into FOUR pairs that would show similar chemical properties, naming the group each pair belongs to. [3] (b) For each pair, state whether the elements are metals, non-metals, or noble gases. [2]	5	OR	Two atoms have the electronic configurations $1s^2 2s^2 2p^4$ and $1s^2 2s^2 2p^6 3s^1$. The ionization energy of one is 2080 kJ/mole and that of the other is 496 kJ/mole. (a) Match each ionization energy with its correct electronic configuration. [3] (b) Give a reason for your choice, referring to shielding effect and the distance of the valence electron from the nucleus. [2]	5
3(iii)	(a) Using electron dot-and-cross diagrams, represent the formation of aluminium fluoride (AlF_3) from an aluminium atom (Z=13) and fluorine atoms (Z=9), showing the electrons transferred and stating how many electrons the aluminium atom loses. [3] (b) An atom with atomic number 16 forms an anion. State the charge on this ion and write its electron configuration. [2]	5	OR	(a) Represent the formation of cations for strontium (Sr, Z=38) and barium (Ba, Z=56) using electron-dot diagrams, stating the charge formed on each. [3] (b) Silicon (Group IVA) has 4 valence electrons. State whether it is more likely to form ionic or covalent bonds with other non-metals, and explain why using its position in the periodic table. [2]	5
3(iv)	Indigo, the dye used to colour blue jeans, has the molecular formula $\text{C}_{16}\text{H}_{10}\text{N}_2\text{O}_2$. It is derived from a compound called indoxyl, $\text{C}_8\text{H}_7\text{ON}$. (a) Calculate the molar mass of indigo, $\text{C}_{16}\text{H}_{10}\text{N}_2\text{O}_2$. (C=12, H=1, N=14, O=16) [2] (b) Calculate the molar mass of indoxyl, $\text{C}_8\text{H}_7\text{ON}$. [1] (c) Write the empirical formula of each compound. [2]	5	OR	A molecule of a phosphorus oxide contains 4 phosphorus atoms and 10 oxygen atoms. (a) Write its molecular formula and empirical formula. [2] (b) Calculate the molar mass of the molecule. (P=31, O=16) [2] (c) Using the rules for writing a molecular formula from a structural formula, write the molecular formula of $\text{CH}_3\text{-CH}_2\text{-OH}$ and $\text{CH}_3\text{-CO-CH}_3$. [1]	5

Invigilator's Signature: _____

Candidate's Name / Roll No.: _____

Chapters Covered: Ch 1 (Nature of Chemistry) | Ch 2 (Matter) | Ch 3 (Atomic Structure) | Ch 4 (Periodic Table) | Ch 5 (Chemical Bonding) | Ch 6 (Stoichiometry) | Ch 7 (Electrochemistry) | Ch 8 (Energetics) | Ch 9 (Chemical Equilibrium)

— End of Question Paper —